Responsible Software. Software that is engineered in a responsible way.
Making engineering design decisions responsibly:
1. With a goal to do good.
2. While preventing avoidable negative impacts.
3. Taking people, other systems, social structures and our planet into account.
Active Responsibility. proactively anticipating and preventing negative impacts before they occur.
we look at the ethical point of view not legal one
Ethics. principles and standards that guide human behavior in various contexts.
Metaethics. "what does it mean for an action to be right ?"
Normative. "how should people act in general (moral principles)?"
Applied. (Our approach) "how can i act ethically in a real-world situation?"

Technical decisions can have **ethical** implications.
Ethical Issue. any situation that may compromise, in whole or in part, the respect of at least one ethical value or principle.
Ethical Agency. capacity to make decisions and take actions that align with ethical values and principles.
Ethical Sensitivty. ability to
- recognize ethical issues (awareness)
- identify their impact: who, what

Ethical Decision Making.
Recognize(knowledge+experience+habits) → Evaluate(cognitive+reason+affective) → Act.
Ethical Lenses.
1. **Desirability.** The product corresponds to what people want.
2. **Viability.** The product generates profits for a business.
3. **Feasibility.** The product can be created with existing or new technology.

Fourth criterion (Isaac et al.). Add **Ethicality**: the product should also be ethical.

Ethical Dilemmas. decision between two or more alternatives, which all have conflicts with ethical values or principles.
Genuine ethical dilemmas are dilemmas in which none of the options override the other, wrong-wrong choice.
Trolley problem.
thought experiment where you must choose between doing nothing and letting a runaway trolley kill several people, or intervening to divert it so fewer people die, testing which moral principles justify action versus inaction.
Moral side.
Emotions should be used explicitely/intentionally for responsible design. (Roeser)
Moral emotions = experienced in response to moral issues or motivating immoral behavior.
Cognitive side.
- **Utilitarianism** Consequences: choose the greatest good for greatest number
- **Deontology** Action: do the right thing according to universal principles
- **Virtue** Person: act as a virtuous person
- **Care** Relationships: mutual responsibility and care for each other

---

Stakeholder Individuals, groups, organizations, institutions, and societies that might reasonably be affected by the technology.
Stakeholder analysis. Identifying every entity that can be affected by the software positively/negatively.
1. Brainstorm to find all stakeholders possible using questions.
2. Use Direct vs Indirect Chart to find new stakeholders.

Direct Stakeholder. in contact with the product.
Indirect Stakeholder. affected through consequences of the usage of the software.
Additional Methodologies: *Ethix. add unintended stakeholders*, *Mitchell. add attributes to stakeholders*, *SAS. use different chart with scale for impact from product and influence on the product.* **Ethical Canvas.** Structure stakeholder, impact, and mitigation info systematically. **Ethical Speculation.** How to think and not what to think - anticipate potential impacts on stakeholders. Create a fictional story in two parts. 1. Check pessimistic outcomes *Black Mirror.* 2. Positive resolutions and future scenarios.